Introduction to Abstract Algebra III: Homework 1

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Exercise 39.2. Show that given $\alpha = \sqrt{2} + \sqrt{3} \in \mathbb{C}$ is algebraic over $\mathbb{Q}$ by finding $f(x) \in \mathbb{Q}[x]$ such that $f(\alpha) = 0$.

Solution. We want to find $f(x) \in \mathbb{C}[x]$ such that $f(\alpha) = 0$. Squaring α, we get $\alpha^2 = 2 + 3 + 2\sqrt{6} = 5 + 2\sqrt{6}$. Then, $(\alpha - 5)^2 = (2\sqrt{6})^2 = 24$. Thus, we have $(\alpha - 5)^2 - 24 = 0$. Rearranging gives us $\alpha^2 - 10\alpha + 1 = 0$. Thus, we can take $f(x) = x^2 - 10x + 1$. □

Exercise 39.8. Find $\operatorname{irr}(\alpha, \mathbb{Q})$ and $\deg(\alpha, \mathbb{Q})$ given the algebraic number $\alpha = \sqrt{2} + i \in \mathbb{C}$. Be prepared to prove that your polynomials are irreducible over $\mathbb{Q}$ if challenged to do so.

Solution. Squaring α, we get $\alpha^2 = 1 + 2\sqrt{2}i$. Then, we have $(\alpha^2 - 1)^2 = -8$. Rearranging gives us $\alpha^4 - 2\alpha^2 + 9 = 0$. Thus, we can take $f(x) = x^4 - 2x^2 + 9$. Lastly, we want to show that $f(x)$ is irreducible. Say f has linear factor $x - \beta$. Then, the root β must divide 9, so $\beta = \pm 1$, ± 3, or ± 9. But, evaluating the polynomial at all those points gives us a non-zero value, so f has no linear factor. Now, say that f can be written as the product of two quadratics, $f(x) = (x^2 + ax + b)(x^2 + cx + d)$. Expanding, we get:

$$f(x) = (x^2 + ax + b)(x^2 + cx + d) = x^4 + (a + d)x^3 + (e + b + ab)x^2 + (ae + bd)x + be.$$

We have $be = 9$, $b + e = \pm 10$ or ± 6. From this, we get the contradiction that $a = \sqrt{3}$ or $a = -\sqrt{3}$, so f has no quadratic factorization. Thus, f is irreducible over $\mathbb{Q}$, so $\operatorname{irr}(\alpha, \mathbb{Q}) = f(x) = x^4 - 2x^2 + 9$ and $\deg(\alpha, \mathbb{Q}) = 4$. □

Exercise 39.10. Classify $\alpha = 1 + i$ as algebraic or transcendental over the field $F = \mathbb{R}$. If α is algebraic over F, find $\deg(\alpha, F)$.

Solution. The number α is algebraic over $\mathbb{R}$, since it is a zero of a polynomial, $f(x) \in \mathbb{R}[x]$. To find this polynomial, we square α to get $\alpha^2 = 2i$. Squaring again gives $\alpha^4 = -4$. Thus, we have $\alpha^4 + 4 = 0$, so $f(x) = x^4 + 4$ is a polynomial in $\mathbb{R}[x]$ such that $f(\alpha) = 0$. Notice that $f(x)$ is reducible over $\mathbb{R}$, specifically, we have:

$$f(x) = \left(x^2 + 2x + 2\right)\left(x^2 - 2x + 2\right) = p(x)q(x).$$

Notice that α is a root of $q(x)$, so $\operatorname{irr}(\alpha, \mathbb{R}) = q(x) = x^2 - 2x + 2$ and $\deg(\alpha, \mathbb{R}) = 2$. □

Exercise 39.16. Classify $\alpha = \pi^2$ as algebraic or transcendental over the field $F = \mathbb{Q}(\pi^3)$. If α is algebraic over F, find $\deg(\alpha, F)$.

Solution. Say that there exists a vanishing polynomial $f(x) = a_0 + a_1x^1 + \cdots + a_nx^n \in \mathbb{Q}[x]$ such that $f(\pi^2) = 0$. Then, $f(\pi^2) = a_0 + a_1\pi^2 + a_2\pi^4 + \cdots + a_n\pi^{2n} = 0$. This implies that for $g(x) = a_0 + a_1x^2 + \cdots + a_nx^n$, we have $g(\pi) = 0$. This contradicts the fact that π is transcendental over $\mathbb{Q}$. Thus, there is no vanishing polynomial for α, so α is transcendental over F. □

Exercise 39.17. Refer to Example 39.20 of the text. The polynomial $x^2 + x + 1$ has a zero α in $\mathbb{Z}_2(\alpha)$ and thus must factor into a product of linear factors in $(\mathbb{Z}_2(\alpha))[x]$. Find this factorization. [*Hint:* Divide $x^2 + x + 1$ by $x - \alpha$ by long division, using the fact that $\alpha^2 = \alpha + 1$.]

Solution. We first divide $x^2 + x + 1$ by $x - \alpha$ using long division. Multiplying $x - \alpha$ by x gives us $x^2 - \alpha x$. Subtracting gives us $x + 1 + \alpha x = (1 + \alpha)x + 1$. Next, we multiply $x - \alpha$ by $1 + \alpha$ to get $(1 + \alpha)x + (1 + \alpha)\alpha$. Subtracting gives us $1 - (1 + \alpha)\alpha = 0$. Thus, we have the factorization $x^2 + x + 1 = (x - \alpha)(x + 1 + \alpha)$ in $(\mathbb{Z}_2(\alpha))[x]$. □

Exercise 39.18.

(i) Show that the polynomial $x^2 + 1$ is irreducible in $\mathbb{Z}_3[x]$.

(ii) Let α be a zero of x^2+1 is an extension of the field $\mathbb{Z}_3$. As in Example 39.20, give the multiplication and addition tables for the nine elements of $\mathbb{Z}_3(\alpha)$, written in order 0, 1, 2, α, 2α, $1+\alpha$, $1+2\alpha$, $2+\alpha$, and $2+2\alpha$.

Solution to (i). To show that x^2+1 is irreducible over $\mathbb{Z}_3[x]$, we can check if it has any roots in $\mathbb{Z}_3$. Evaluating x^2+1 at 0, 1, and 2 gives us:

$$(0)^2+1=1, \quad (1)^2+1=2, \quad \text{and} \quad (2)^2+1=5\equiv 2 \pmod 3.$$

Thus, x^2+1 has no roots in $\mathbb{Z}_3$, so it is irreducible over $\mathbb{Z}_3[x]$. □

Solution to (ii).

$+$	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
0	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
1	1	2	0	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$	α	2α
2	2	0	1	$2+\alpha$	$2+2\alpha$	α	2α	$1+\alpha$	$1+2\alpha$
α	α	$1+\alpha$	$2+\alpha$	2α	0	$1+2\alpha$	1	$2+2\alpha$	2
2α	2α	$1+2\alpha$	$2+2\alpha$	0	α	1	$1+\alpha$	2	$2+\alpha$
$1+\alpha$	$1+\alpha$	$2+\alpha$	α	$1+2\alpha$	1	$2+2\alpha$	2	2α	0
$1+2\alpha$	$1+2\alpha$	$2+2\alpha$	2α	1	$1+\alpha$	2	$2+\alpha$	0	α
$2+\alpha$	$2+\alpha$	α	$1+\alpha$	$2+2\alpha$	2	2α	0	$1+2\alpha$	1
$2+2\alpha$	$2+2\alpha$	2α	$1+2\alpha$	2	$2+\alpha$	0	α	1	$1+\alpha$

Table I: Addition table for $\mathbb{Z}_3(\alpha)$

$\times$	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
0	0	0	0	0	0	0	0	0	0
1	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
2	0	2	1	2α	α	$2+2\alpha$	$2+\alpha$	$1+2\alpha$	$1+\alpha$
α	0	α	2α	2	1	$2+\alpha$	$1+\alpha$	$2+2\alpha$	$1+2\alpha$
2α	0	2α	α	1	2	$1+2\alpha$	$2+2\alpha$	$1+\alpha$	$2+\alpha$
$1+\alpha$	0	$1+\alpha$	$2+2\alpha$	$2+\alpha$	$1+2\alpha$	2α	2	0	α
$1+2\alpha$	0	$1+2\alpha$	$2+\alpha$	$1+\alpha$	$2+2\alpha$	2	α	2α	1
$2+\alpha$	0	$2+\alpha$	$1+2\alpha$	$2+2\alpha$	$1+\alpha$	0	2α	α	2
$2+2\alpha$	0	$2+2\alpha$	$1+\alpha$	$1+2\alpha$	$2+\alpha$	α	1	2	2α

Table II: Multiplication table for $\mathbb{Z}_3(\alpha)$

These tables were generated using the fact that $\alpha^2 = -1$ and the distributive property of multiplication over addition. $\square$

Exercise 39.24. We have stated without proof that π and e are transcendental over $\mathbb{Q}$.

(i) Find a subfield F of $\mathbb{R}$ such that π is algebraic of degree 3 over F.

(ii) Find a subfield E of $\mathbb{R}$ such that e^2 is algebraic of degree 5 over E.

Solution to (i). Take the vanishing polynomial $f(x) = x^3 - \pi^3$. We now, have to show irreducibility. Assume that f has a linear factor $x - \beta$. Then $\beta \in \mathbb{Q}$ must divide π^3. But if $k\beta = \pi^3$, then π^3 would be rational, which is a contradiction. Thus, there is no linear factor. Since f has degree 3, it cannot have a quadratic factorization. Thus, f is irreducible over $\mathbb{Q}$, so we can take $F = \mathbb{Q}(\pi) = \mathbb{Q}[x]/\langle f(x)\rangle \subset \mathbb{R}$, such that π is algebraic of degree 3 over F. $\square$

Solution to (ii). Take the vanishing polynomial $g(x) = x^5 e^{10}$. We now, have to show irreducibility. Assume that g has a linear factor $x - \beta$. The same logic holds from part (i) to show that β cannot be rational. Now, assume that g has a quadratic or cubic factorization. Then, we can write $g(x) = (x^2+ax+b)(x^3+cx^2+dx+e)$. Then $be = e^{10}$, which implies that either b or e is irrational. Thus, there is no quadratic or cubic factorization. Thus, we can take $E = \mathbb{Q}(e) = \mathbb{Q}[x]/\langle g(x)\rangle \subset \mathbb{R}$, such that e^2 is algebraic of degree 5 over E. $\square$

Exercise 39.29. Let E be an extension field of F, and let $\alpha, \beta \in E$. Suppose α is transcendental over F but algebraic over $F(\beta)$. Show that β is algebraic over $F(\alpha)$.

Solution. Let $f(x) = \operatorname{irr}(\alpha, F(\beta)) = a_n x^n + \cdots a_1 x^1 + a_0$. Notice that $f(\alpha) = 0$, so we have $a_n\alpha^n + \cdots + a_1\alpha^1 + a_0 = 0$. Rearranging gives us $a_n\alpha^n + \cdots + a_1\alpha^1 = -a_0$. Since α is transcendental over F, we can view this as a polynomial in β with coefficients in $F(\alpha)$. Thus, we have $g(\beta) = a_n\alpha^n + \cdots + a_1\alpha^1 + a_0 = 0$, where $g(x) \in F(\alpha)[x]$. This implies that β is algebraic over $F(\alpha)$. $\square$

Exercise 39.30. Let E be an extension field of a finite field F, where F is q elements. Let $\alpha \in E$ be algebraic over F of degree n. Prove that $F(\alpha)$ has q^n elements.

Solution. Let $\operatorname{irr}(\alpha, F) = a_n x^n + \cdots + a_1 x^1 + a_0$. Notice that $F(\alpha) = \{a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha^1 + a_0 : a_i \in F\}$, since $\operatorname{irr}(\alpha, F)$ is irreducible over F. Since F has q elements, there are q choices for each of the coefficients a_i. Thus, there are q^n elements in $F(\alpha)$. $\square$

Exercise 39.32. Consider the prime field $\mathbb{Z}_p$ of characteristic $p \neq 0$.

(i) Show that, for $p \neq 2$, not every element in $\mathbb{Z}_p$ is a square of an element of $\mathbb{Z}_p$. [*Hint:* $1^2 = (p-1)^2 = 1$ in $\mathbb{Z}_p$. Deduce the desired conclusion by *counting*.]

(ii) Using part (i), show that there exist finite fields of p^2 elements for every prime p in $\mathbb{Z}^+$.

Solution to (i). By the hint, we already know that 1 is a square in $\mathbb{Z}_p$. Since $p \neq 2$, we have $p - 1 \neq 1$, so $p - 1$ is not a square in $\mathbb{Z}_p$. Thus, there are at least two elements in $\mathbb{Z}_p$ that are not squares. Since $\mathbb{Z}_p$ has p elements, there are at most $p - 2$ squares in $\mathbb{Z}_p$. Thus, not every element in $\mathbb{Z}_p$ is a square of an element of $\mathbb{Z}_p$. $\square$

Solution to (ii). For each prime $p \in \mathbb{Z}^+$, consider any irreducible polynomial $f(x) \in \mathbb{Z}_p[x]$ of degree p. Such a polynomial exists by part (i). Then, we know that there exists a field extension E of $\mathbb{Z}_p$ containing a zero α of f. Then, $\mathbb{Z}_p(\alpha)$ has elements of the form $a + b\alpha, a, b \in \mathbb{Z}_p$. Since $\mathbb{Z}_p$ has p elements, there are p^2 such elements in $\mathbb{Z}_p(\alpha)$. Thus, there exist finite fields of p^2 elements for every prime p in $\mathbb{Z}^+$. $\square$